

Anutej Sachin Kardele

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EXPERIENCE

Software Engineering Practicum, HeinOnline, Buffalo, NY

Jan 2026 – May 2026

- Engineered a Python-powered OCR and AI extraction pipeline using Qwen LLMs, performing ablation studies to optimize multi-document processing; achieved 90.58% extraction accuracy across 28,167 journal pages.
- Built metadata verification services matching authors against PostgreSQL records and enriching metadata with Crossref where available (30%); used the AI pipeline for remaining documents, cutting manual data entry by 90%.

Software Engineer, Simbott Technologies, Mumbai, India

Mar 2024 – Jul 2025

- Architected a reusable, component-based C# framework for enterprise applications, reducing duplicated code by 80% and enabling custom module implementation in under 5 minutes.
- Co-engineered 3 custom peripherals in collaboration with the electronics team, authoring high-performance firmware to establish low-latency serial communication pipelines between edge hardware and the central simulation engine.
- Engineered real-time telemetry processing pipelines for custom physical controllers, optimizing sensor data streams to decrease average system latency by 68% in air-gapped environments.

Software Developer, Tecknotrove System Pvt Ltd, Mumbai, India

Jun 2022 – Nov 2023

- Implemented core C# application logic for enterprise training software, integrating a MySQL relational database to securely capture, store, and query thousands of user evaluation data points in real time.
- Engineered a scoring algorithm that extracts UV texture coordinates via vector raycasting and replaced legacy physics colliders with 2D color-map sampling, cutting CPU workload by 86% and eliminating clipping errors.
- Executed a full-scale core engine migration (Unity 5.6 to 2019) for a legacy enterprise project, resolving 54 critical package dependencies and establishing strict hardware-to-software interfacing protocols.

EDUCATION

M.S. in Computer Science, University at Buffalo, SUNY | **GPA: 3.75 / 4.0**

Aug 2025 – Dec 2026

B.Tech. in Computer Science, Vellore Institute of Technology, India | **CGPA: 7.72 / 10.0**

Jul 2018 – May 2022

PROJECTS

OpenStream | Java, Spring Boot, React, PostgreSQL, JUnit 5, Mockito, Docker | openstream.anutej.us

Sep 2026

- Built and deployed a full-stack microblogging platform with React and an 11-endpoint Spring Boot REST API using Spring Data JPA, Hibernate, PostgreSQL on Neon, and a layered service architecture.
- Integrated the BERTweet Guard FastAPI microservice on Google Cloud Run via Spring RestClient; rejected flagged posts with HTTP 422 and handled moderation timeouts and service outages.
- Modeled self-referential follow relationships and optimized feed retrieval with JPQL JOIN FETCH to prevent N+1 queries; expanded to 17 JUnit 5/Mockito tests and deployed the Dockerized backend to Render.

BERTweet Guard | Python, PyTorch, Hugging Face, LoRA, FastAPI, Docker | bertweet.anutej.us

Jan 2026 – May 2026

- Fine-tuned BERTweet with LoRA (r=32) on 217,270 deduplicated examples from four NLP datasets; used per-source splits and source/class-weighted sampling for cross-domain moderation.
- Built source-balanced checkpoint selection and threshold calibration using Macro F1, achieving 0.9054 Test Macro F1, 0.8309 source-balanced Macro F1, and 93.01% accuracy.
- Merged LoRA adapters and exported model artifacts to Hugging Face; containerized a FastAPI service with Docker and deployed to Google Cloud Run via Cloud Build for OpenStream moderation.

Interactive Portfolio | React, Vite, Spring Boot, Gemini API, GitHub Actions | anutej.us

Feb 2026

- Built a React/Vite SPA replicating the VS Code interface, including an in-memory virtual file system, responsive component rendering, dynamic tabs, and light/dark themes.
- Developed a terminal emulator with a custom lexical parser supporting UNIX-style commands such as ls, cd, and open to navigate the virtual file system.
- Built a Spring Boot REST proxy for the Gemini API with server-side credential management, per-IP rate limiting, and retry handling; automated frontend CI/CD to GitHub Pages with GitHub Actions.

SKILLS

- **Languages:** Java, Python, C#, JavaScript, SQL, Go, C, C++, HTML/CSS
- **Backend & Data:** Spring Boot, Spring Data JPA, Hibernate, FastAPI, REST APIs, PostgreSQL, MySQL
- **Frontend:** React, Vite, Tailwind CSS, React Router
- **Cloud & DevOps:** Docker, Google Cloud (Cloud Run, Cloud Build), GitHub Actions, CI/CD, Git, Maven, Linux/Unix
- **Testing & AI/ML:** JUnit 5, Mockito, Machine Learning, PyTorch, Hugging Face Transformers, LoRA, scikit-learn
- **Core Concepts:** OOP, Data Structures & Algorithms, Design Patterns, SOLID, Concurrency, Distributed Systems

HONORS & AWARDS

National Winner, Smart India Hackathon 2020 | VR Fire Safety Simulator

Dec 2020